



Solubility of CO₂ in Glassy PMMA and PS over a Wide Pressure Range: The Effect of Carbonyl Groups

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Abstract

The equilibrium sorption isotherms of CO₂ in glassy PMMA and PS at 32, 42, and 52 °C over a wide pressure range from 20 to 340 atm were investigated. The normalized sorption concentration (C) isotherms for the polymers in terms of pressure (P) can be described fairly well by two power laws $C = kP^n$ for below and above critical pressure (P_c). The exponent n is a measure of sorption intensity and is closely associated with the interaction between CO₂ and the polymer. From the temperature variation of n values, a negligible interaction between CO₂ and the polymer is found in the sorption process below P_c whereas the interaction is significant above P_c . For a given temperature, the n value for PMMA is 1~2 times higher than that for PS as a result of the specific interaction of CO₂ and the carbonyl groups in PMMA. The pre-exponential constant k is a measure of sorption capacity and is closely associated with the heat of sorption. In sorption above P_c , k is found to decrease with increasing temperature due to the exothermic sorption process that leads to a decrease in sorption capacity with temperature. From the sorption isobars of PMMA and PS, we observe that the temperature necessary for chemisorption to occur is lower for PMMA than for PS; this results from the specific interaction of CO₂ and the carbonyl groups of PMMA.

Introduction

The carbon dioxide (CO₂) sorption isotherms in glassy polymers exhibit a downward curvature as the isothermal sorption concentrations are plotted versus CO₂ pressures below the critical pressure. These isotherms have been reported to follow the dual-mode sorption model [1–11] in which the total sorption concentration C is expressed by a combination of a void-filling Langmuir-type sorption C_H for low pressures and a Henry's law dissolution C_D for high pressures as described by Eq. (1). In terms of CO₂ pressure p , the dual-mode model is written as in Eq. (2):

$$C = C_H + C_D, \quad (1)$$

$$C = \frac{C'_H bp}{1 + bp} + k_D p, \quad (2)$$

where C'_H may be interpreted as a site capacity constant for the Langmuir sorption with the parameter b describing the affinity of the gas for these sites, while k_D represents the Henry's law parameter and can be interpreted as the ordinary solubility constant [5]. In Eq. (2), the Langmuir sorption term is reduced to a pressure independent C'_H for high pressures, whereas the value of the Henry's law term increases linearly with increasing pressures. The C'_H is zero as the contribution of the Langmuir sorption disappears.

Koros and Paul [3] reported that the total CO₂ sorption concentration in poly(ethylene terephthalate) (PET) at above T_g over a pressure range 0–20 atm followed Henry's law while the dual-mode sorption model applied in the glassy state. The CO₂ sorption isotherms for cellulose acetate at pressures up to 45 atm followed also the dual-mode sorption model at below T_g of the polymer [11]. In Berens and Huvard's work [7], a dual-mode CO₂ sorption isotherm (the curvature was concave downward) was shown to be characteristic of the glassy state of a polymer, but the curvature was upward for the rubbery state over a pressure range 0–65 atm. In Kamiya and coworkers' work [8], a different type of CO₂ sorption isotherm was observed at below T_g of poly(vinyl benzoate). This different type of isotherm was concave to the pressure axis in the low-pressure region and turned into a straight line with increasing CO₂ pressure (up to 50 atm) that could be extrapolated back to the coordinate origin. According to the reports described above, neither model was a universal model, implying that the nature of the polymer and/or the pressure range affect the sorption isotherms. In addition, CO₂ is not an ideal gas (particularly true at high pressures) and its properties (such as density, viscosity, intermolecular attraction force, and diffusivity) are not linearly proportional to pressures. These CO₂ properties (especially the density) are even considerably varied with

pressures near the critical conditions (72.8 atm and 31.1 °C). The CO₂ sorption isotherms for high pressures crossing the critical pressure might, thus, deviate from the Henry's law dissolution model. Furthermore, this deviation might as well arise from the affinity of CO₂, which varies considerably depending on the type of polymer of interest.

The affinity of CO₂ for a polymer is associated with the interaction between CO₂ and the polymer. Fourier transform infrared (FTIR) spectroscopy has been used to investigate the interactions of CO₂ with polymers [12–15]. In Eckert's findings [15], polymers containing carbonyl groups act as an electron donor and exhibit a specific intermolecular interaction with CO₂ acting as an electron acceptor rather than as an electron donor. Johnston and coworkers [16] have also suggested that the interaction of CO₂ with polymers possessing acrylate groups (containing carbonyl groups) may be of a Lewis acid-base nature. Specific interactions were proposed to exist between CO₂ and the dipoles of C–F bonds [17] or fluorine [18] to explain the increased solubility of CO₂ in fluorine containing polymers. Weak electrostatic interactions [15, 19] were proposed to exist between CO₂ and the π -system (phenyl ring) in PS. PS does not possess carbonyl groups and thus does not show strong Lewis acid-base interaction with CO₂. Therefore, the dissolution behavior of CO₂ in a polymer might vary considerably with the nature of the polymer, and the Henry law used to describe the dissolution behavior at high pressures might no longer be applicable.

For many glassy polymers, the CO₂ sorption isotherms reported were measured in a pressure range between 0 and up to 30 atm; the Langmuir capacity contribution was clearly observed at pressures less than about 5 atm depending on temperatures [2, 3]. A Langmuir-type sorption is small as compared with the dissolution in a bulk polymer for high pressures [20]. The contribution of the Langmuir-type sorption isotherms becomes increasingly negligible as the pressures increase for the glassy polymers and it disappears entirely for the rubbery polymers. The objective of this study is to measure the CO₂ sorption isotherms and isobars for glassy PMMA and PS over a wide pressure range of 20 to 340 atm. The applicability of the Henry law dissolution model for sorption concentration isotherms in terms of pressures above 30 atm is investigated specifically. The effects of the carbonyl groups in PMMA and the phenyl groups in PS on the CO₂ sorption isotherms and isobars are focused upon.

Experimental

Materials

Poly(methyl methacrylate) (PMMA) having $M_w = 396\,000$ and $M_w/M_n = 3.72$ was supplied by Chi Mei Corporation (Tainan, Taiwan). Polystyrene (PS) having a melt index 5.0 was also supplied by Chi Mei Corporation with a product number PH-88. Both polymers were used as received in the form of sheets. The samples used for sorption measurements were ~ 1 cm wide, ~ 3 cm long, and ~ 0.15 cm thick.

Sorption Measurements

The weight of a sample for the sorption measurement was ~ 2.5 g. The sorption measurements were performed in a supercritical extractor supplied by ISCO [Lincoln, Nebraska], model SFX 2-10. The samples for the sorption measurements were put in a 10-cm³ cell located inside the extractor and pressurized by the supplied syringe-type pump, model 260D. The extractor could control pressure over a range of 0–10000 psi (680 atm) and a temperature over a range of 20–100 °C. The pressurizing time for reaching any preset pressure below 680 atm was 30 sec. The extractor was controlled at a preset temperature prior to the pressurizing. The exposure (soaking) time for the sample in CO₂ was 1 hour, which was experimentally confirmed to be long enough for the rectangular samples to reach equilibrium sorption concentration as shown in Figure 1. After exposure, the cell was depressurized to ambient pressure in 10 seconds and evacuated for weight measurements of samples in another 10 seconds. The data for weight change were determined to be highly reproducible. The weight changes were measured outside the cell, not as measured *in situ* inside the cell, because the measurements were good enough to provide the equilibrium dissolution isotherms for the purpose of this study. After the measurements in weight changes, the samples were heated at 110 °C and vacuumed for 30 min to evacuate all dissolved CO₂ inside the sample. Such heating treatments demonstrated that the samples after sorption experiments showed negligible polymer dissolution in CO₂.

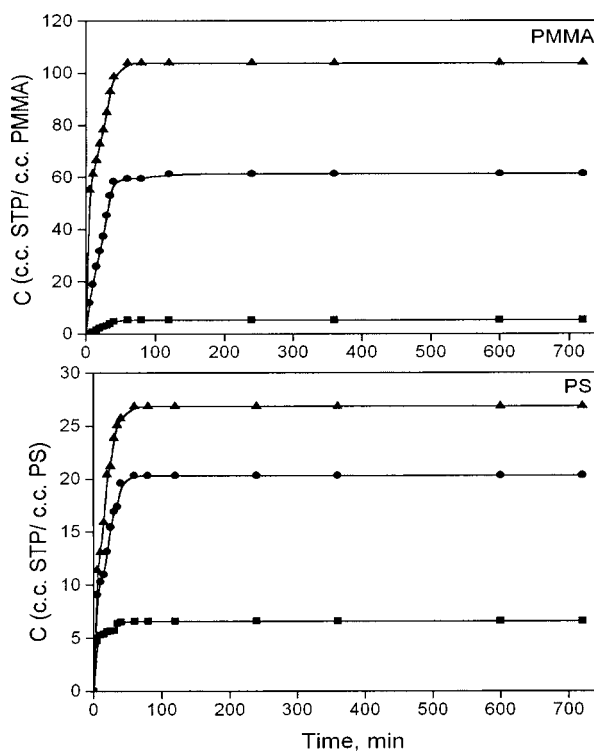


Figure 1. The CO₂ sorption concentration in PMMA and PS as a function of time for 34 atm (■), 72.8 atm (●), and 204 atm (▲).

Results and Discussion

As described in the experimental section, sheet samples were used for sorption measurements in this study; they were ~1.5 mm in thickness, which is much thicker than the film samples used previously [1–11, 22]. The purpose of this study is to investigate the equilibrium sorption isotherms through which the dissolution behavior of CO₂ in PMMA and PS can be compared and correlated with the CO₂ conditions. The equilibrium sorption, or more appropriately equilibrium solubility, refers to the saturated dissolution of CO₂ in the polymers. For sheet samples in this study, it takes time (at least 60 min, as will be demonstrated later) to reach equilibrium sorption and to release the dissolved CO₂ in the polymers. Provided that the depressurizing and the weighing for the sample after the sorption experiment are complete in a very short time (totally 20 seconds in this study), it is not unreasonable to assume that the weight change obtained from the weighing outside the cell after the sorption experiment represents the total weight change due to the sorption process. Although the data on weight change obtained are not from the *in-situ* weighing during the sorption process in this study, the data were confirmed to be highly reproducible and were good enough to provide the equilibrium dissolution isotherms for the purpose of this study.

Figure 1 shows the CO₂ sorption isotherms at 32 °C for PMMA and PS samples as a function of exposure time for three pressures. The curves in Figure 1 clearly level off after 60 minutes of exposure, suggesting that the equilibrium solubility is reached at 60 minutes of exposure for both polymers. In the initial 60 minutes exposure, the CO₂ uptake rate increases with increasing pressures. After 60 minutes of exposure, the leveled off sorption concentration at any pressure is much higher for PMMA than for PS. Since PMMA and PS exhibit similar glass transition temperatures near 100 °C, the carbonyl groups in PMMA apparently exhibit a much higher sorption ability than the phenyl groups in PS, leading to a considerable difference in equilibrium sorption concentration between these two polymers. Based on the observation from Figure 1, 60 minutes of exposure time is thus used to perform all the sorption experiments to obtain equilibrium sorption isotherms for this study.

Figures 2 and 3 show the equilibrium sorption concentration of CO₂ as a function of pressure at 32, 42, and 52 °C for PMMA and PS, respectively. The sorption isotherm at each temperature in terms of pressure can be roughly divided into two regions, with the critical pressure roughly at the border of the two regions. The curve of each sorption isotherm for both polymers is upward concave below the critical pressure, whereas the curve is downward concave above the critical pressure. For any given temperature and pressure, PMMA yields a higher sorption concentration than PS does, indicating that the carbonyl groups in PMMA exhibit higher sorption ability than the phenyl groups in PS. These sorption isotherms can be fitted fairly well by two straight lines for the two divided regions, providing that the logarithmic dissolved-CO₂ concentration is plotted as a function of the logarithmic CO₂ pressure, as shown in Figures 4 and 5 for

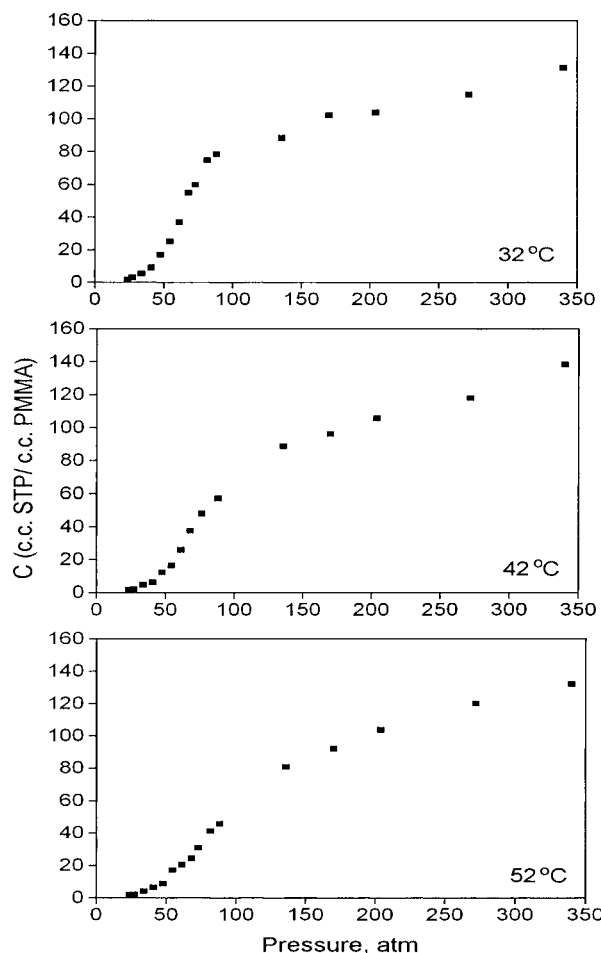


Figure 2. The CO₂ sorption isotherms at 32, 42, and 52 °C for PMMA as a function of CO₂ pressure.

PMMA and PS, respectively. The dissolved-CO₂ concentration for the y axis in Figures 4 and 5 has been normalized on the basis of the content of the carbonyl groups in PMMA (Figure 4) and the content of the phenyl groups in PS (Figure 5) having a unit of cm³ CO₂ (STP) per gram carbonyl group and per gram phenyl group, respectively. This normalization treatment facilitates the examination of the effects of the carbonyl groups and the phenyl groups on the sorption behavior.

Figures 4 and 5 are apparently better than Figures 2 and 3 at demonstrating the two different sorption behaviors for PMMA and PS, respectively, over a wide pressure range. The sorption behavior for both polymers is found to change at near the critical pressure. In other words, the sorption behavior for both polymers is different for below and above the critical pressure. Based on the plots in Figures 4 and 5, the two sorption behaviors for both polymers can be better described by two power laws written as in Eqs (3) and (4) than by the dual-mode model as in Eq. (2):

$$C = k_a P^{n_a}, \quad (3)$$

$$C = k_b P^{n_b}, \quad (4)$$

where C is the equilibrium sorption concentration in a unit of cm³ (STP) CO₂ per gram carbonyl groups for PMMA (or

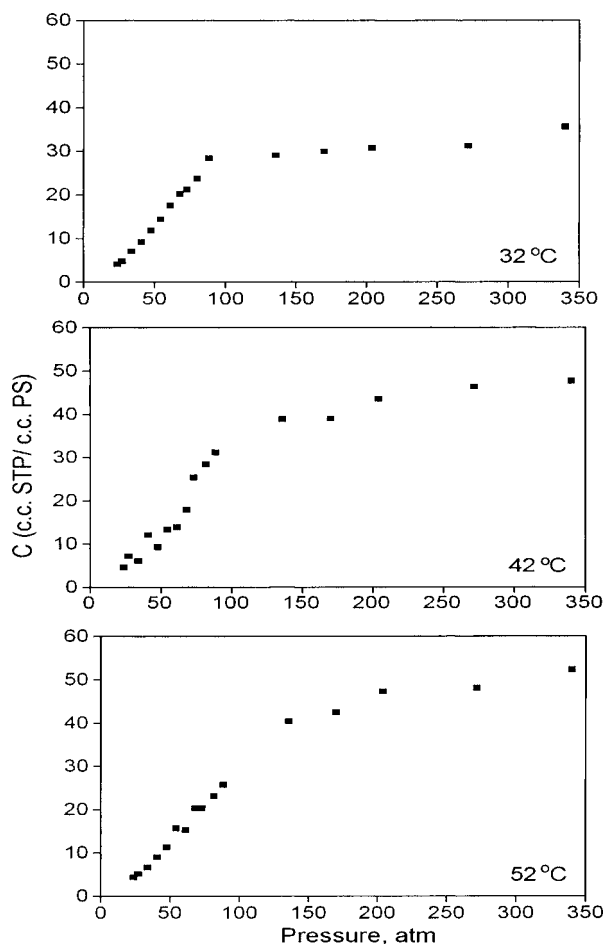


Figure 3. The CO₂ sorption isotherms at 32, 42, and 52 °C for PS as a function of CO₂ pressure.

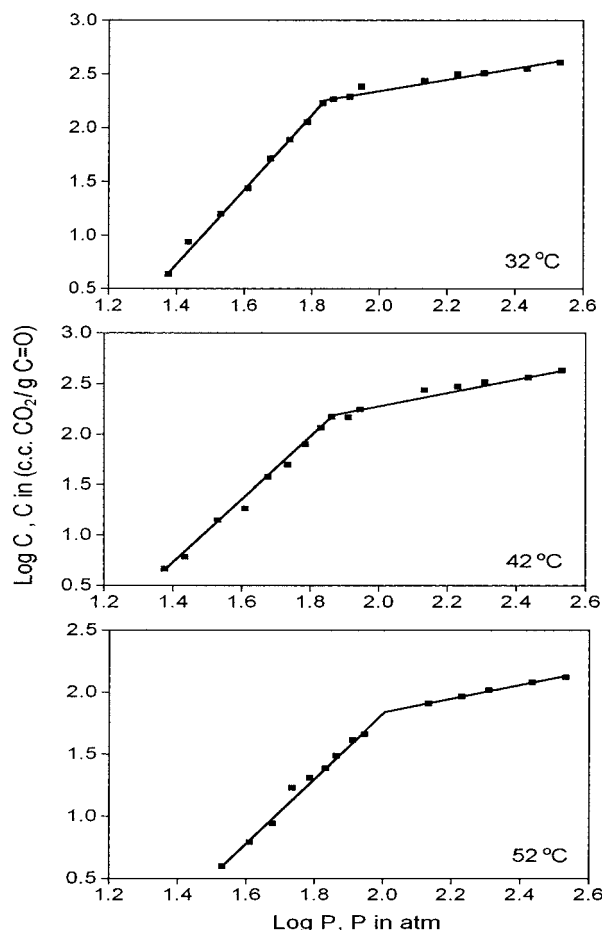


Figure 4. The logarithmic normalized CO₂ sorption isotherms at 32, 42, and 52 °C for PMMA as a function of the logarithmic CO₂ pressure.

per gram phenyl groups for PS), P is the pressure in atm, k_a , k_b , n_a , and n_b are constants. Eq. (3) describes the sorption behavior for above the critical pressure whereas Eq. (4) describes it for below the critical pressure. Apparently, both Eqs (3) and (4) do not include the contribution from either the Langmuir-type sorption or the Henry's law dissolution. In other words, neither the Langmuir-type sorption mode nor the Henry's law dissolution mode can satisfactorily describe the CO₂ sorption behavior as shown in Figures 4 and 5. According to the adsorption theory, the Langmuir adsorption model turns into a power law adsorption model, or a Freundlich adsorption model (named after the originator), as the adsorption surfaces are no longer homogeneous and the variation in the parameter b in the Langmuir model is attributed entirely to a variation in the heat of adsorption [23]. Thus, it is not unreasonable to propose the power law to describe the dissolution behavior of CO₂ in both polymers because the location inside polymers at which the CO₂ molecule is dissolved can be considered a heterogeneous environment.

The k_a , k_b , n_a , and n_b values calculated from Figures 4 and 5 are tabulated in Tables 1 and 2 for PMMA and PS, respectively. For a polymer, k_a and k_b give a measure of sorption capacity at atmospheric pressure for the sorption behavior above and below critical pressure, respectively. n_a and n_b give a measure of sorption intensity according to the

Freundlich adsorption theory for above and below critical pressure, respectively. The sorption intensity is closely associated with the interaction between the polymer and CO₂. As can be observed from Tables 1 and 2, the sorption capacity constants and sorption intensity constants are clearly dependent upon the type of polymer and temperature. For the low-pressure region for both polymers, although the k_b increases with increasing temperatures, any discussion on the temperature variation of k_b is meaningless because its value is too low. The n_b is higher than unity in value and decreases with increasing temperature, indicative of negative apparent activation energy (-8.7×10^3 and -5.1×10^3 J/mol for PMMA and PS, respectively, as calculated from the data in Tables 1 and 2) for the mechanism of the sorption process in the low-pressure region. For a given temperature, the n_b for PMMA is about twice as high as that for PS, suggesting that the carbonyl groups of PMMA exhibit a higher level of interaction with CO₂ than the phenyl groups of PS. For the high-pressure region for both polymers, the k_a decreases with increasing temperature whereas the n_a increases with increasing temperature. The decrease in k_a with increasing temperature might be attributed to the exothermic sorption that leads to a decrease in sorption amount or capacity with temperature. The temperature variation of sorption capacity of PMMA is more significant than that of PS, apparently resulting from the relatively high level of interaction with CO₂

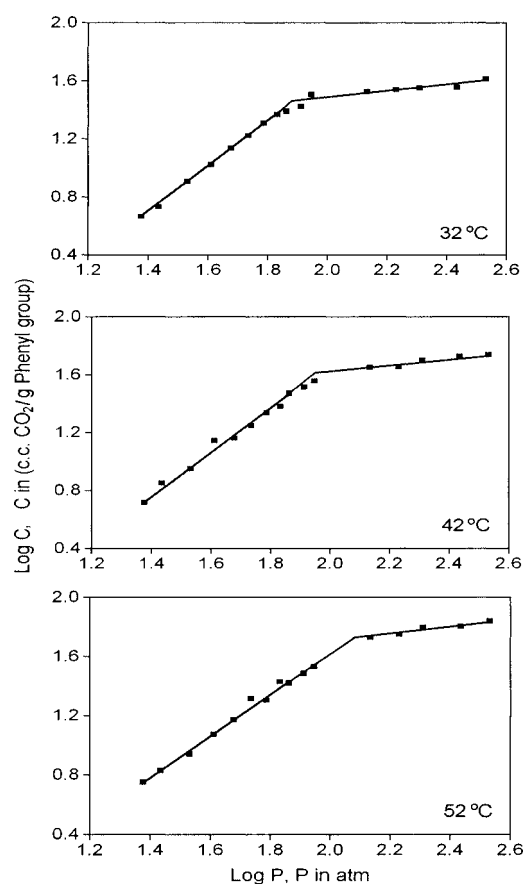


Figure 5. The logarithmic normalized CO₂ sorption isotherms at 32, 42, and 52 °C for PS as a function of the logarithmic CO₂ pressure.

Table 1. Sorption capacity (k_a and k_b) and sorption intensity (n_a and n_b) constants for PMMA

Temperature	Low pressure region		High pressure region	
	k_b	n_b	k_a	n_a
32 °C	1.3×10^{-4}	3.33	30.23	0.44
42 °C	2.2×10^{-4}	3.11	7.66	0.70
52 °C	3.0×10^{-4}	2.69	1.18	0.83

for the carbonyl groups as compared with the phenyl groups. This high level of interaction leads to an exothermic sorption process. Since the temperature variation of n_a increases with increasing temperature, the sorption in the high-pressure region exhibits positive apparent activation energy (2.7×10^4 and 3.4×10^4 J/mol for PMMA and PS, respectively) for the sorption mechanism. For a given temperature, the n_a for PMMA is about 1–2 times higher than that for PS, suggesting again that the carbonyl groups of PMMA exhibit a higher level of interaction with CO₂ than the phenyl groups of PS.

The sorption isotherms that can be described by power laws over a wide pressure range of 20–340 atm are different not only from those described by the dual mode model in Koros and Paul's study [3] over the pressure range 0–~30 atm but also from those in Wissinger and Paulaitis's work [22] over the pressure range 0–100 atm. In Wissinger and Paulaitis's work [22], the CO₂ sorption isotherms for polycarbonate (PC) having a T_g 147 °C, PMMA having a T_g

Table 2. Sorption capacity (k_a and k_b) and sorption intensity (n_a and n_b) constants for PS

Temperature	Low pressure region		High pressure region	
	k_b	n_b	k_a	n_a
32 °C	3.0×10^{-2}	1.58	8.48	0.27
42 °C	5.2×10^{-2}	1.43	5.85	0.40
52 °C	5.5×10^{-2}	1.40	2.13	0.62

105 °C, and PS having a T_g 100 °C at elevated pressures up to 100 atm over a temperature range 33–65 °C were investigated. The sorption behavior of the polymer was found to depend on the occurrence of a glass transition induced by the CO₂ sorption. Two distinct types of sorption isotherms were measured. Without the occurrence of a glass transition for the polymer (e.g., PC at 35 °C, PMMA at 32.7 °C) during the sorption experiments, the sorption isotherms reached limiting values at elevated pressures, whereas the isotherms continued to increase with pressure (upward concave curves were observed) for the polymer (e.g., PMMA at 42.0 and 58.8 °C, PS at 35.0, 50.0, and 65 °C) involving a glass transition. According to Wissinger and Paulaitis's work [22], the PMMA and PS in this study should involve a glass transition during sorption experiments at 42 and 52 °C and the sorption isotherms should increase with increasing pressure at elevated pressures. As shown in Figures 2 and 3, however, PMMA and PS both exhibit a very slow increase in sorption concentration at above critical pressure. The sorption behavior that has been observed in this work is, apparently, not consistent with the results from Wissinger and Paulaitis's work [22]. The disparities between this work and previous studies [3, 22] are most likely due to differences in experimental techniques. In this work, sheet samples were used and the sorption experiments were done at up to 340 atm. In the previous studies, thin films were used and the experiments were done at up to only ~30 [3] or 100 atm [22]. CO₂ at these low pressures has a low potential to drive CO₂ molecules into the polymers and leads to low CO₂ sorption concentrations.

Thus far, the carbonyl groups of PMMA have been demonstrated through sorption isotherms to significantly enhance the sorption of CO₂ in the polymer, as compared with the phenyl groups of PS. Figures 6 and 7 show the sorption isobars for three pressures for PMMA and PS, respectively. The sorption isobars can be used to investigate the sorption types involving physical sorption (physisorption) and chemical sorption (chemisorption). Physisorption usually has very low activation energy and is important for gases below their critical temperatures. Chemisorption usually exhibits relatively high activation energy as compared with physisorption. At low temperatures, chemisorption is thus relatively insignificant as compared with physisorption, whereas at high temperatures, physisorption is relatively insignificant as compared with chemisorption. The amounts sorbed by physisorption and chemisorption can be both decreased with increasing temperature because of the exothermic processes of these two types of sorption. There is no sharp dividing

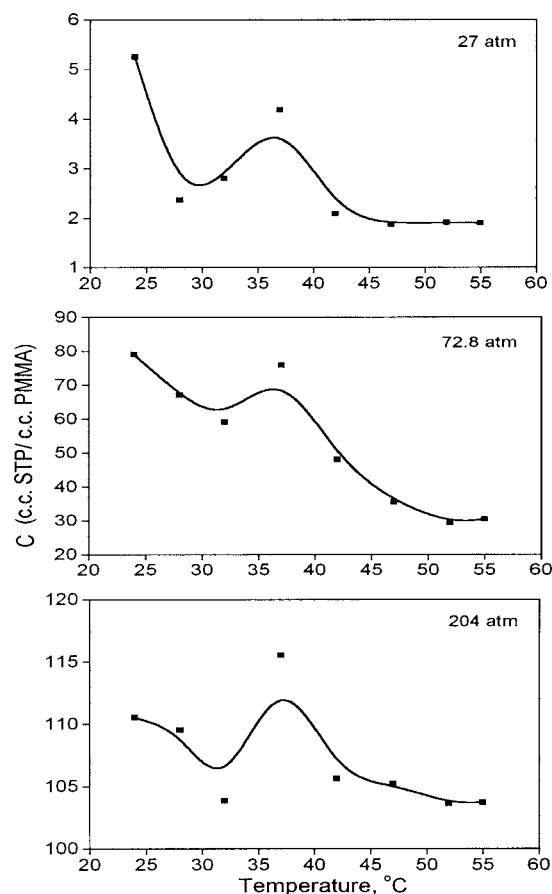


Figure 6. The CO₂ sorption isobars at 27, 72.8, and 204 atm for PMMA as a function of temperature.

line between these two types of sorption. As can be seen in Figure 6, the isobars for PMMA for three pressures at temperatures below $\sim 31^\circ\text{C}$ correspond to physisorptions. The amount sorbed by physisorption is shown to decrease with increasing temperature. In a temperature range of ~ 31 – $\sim 37^\circ\text{C}$, the sorption process is in a transition from the physisorption to the chemisorption, which is clearly seen at $\sim 37^\circ\text{C}$ for the three pressures studied. The amount sorbed by the chemisorption is also shown to decrease with increasing temperature in the same way as for the physisorption. As seen from Figure 7, the physisorption for PS is found to be below 32, 35, and 32°C for pressures of 27, 72.8, and 204 atm, respectively. For the 27 atm, the chemisorption is clearly seen at $\sim 37^\circ\text{C}$ whereas the chemisorption is shifted to $\sim 45^\circ\text{C}$ for the 72.8 atm. For the 204 atm, the transition temperature range from the physisorption to the chemisorption is so broad that the chemisorption is not clearly seen at below 55°C . A comparison between Figures 6 and 7 reveals that the amount sorbed by PMMA is higher than that by PS through either the physisorption or the chemisorption. For any given pressure, the temperature necessary for chemisorption to occur is lower for PMMA than for PS due to the presence of the specific interaction of the carbonyl groups of PMMA with CO₂. This specific interaction results in a relatively low activation energy for the chemisorption for PMMA and leads to the occurrence of chemisorption at a relatively low temperature as compared with PS.

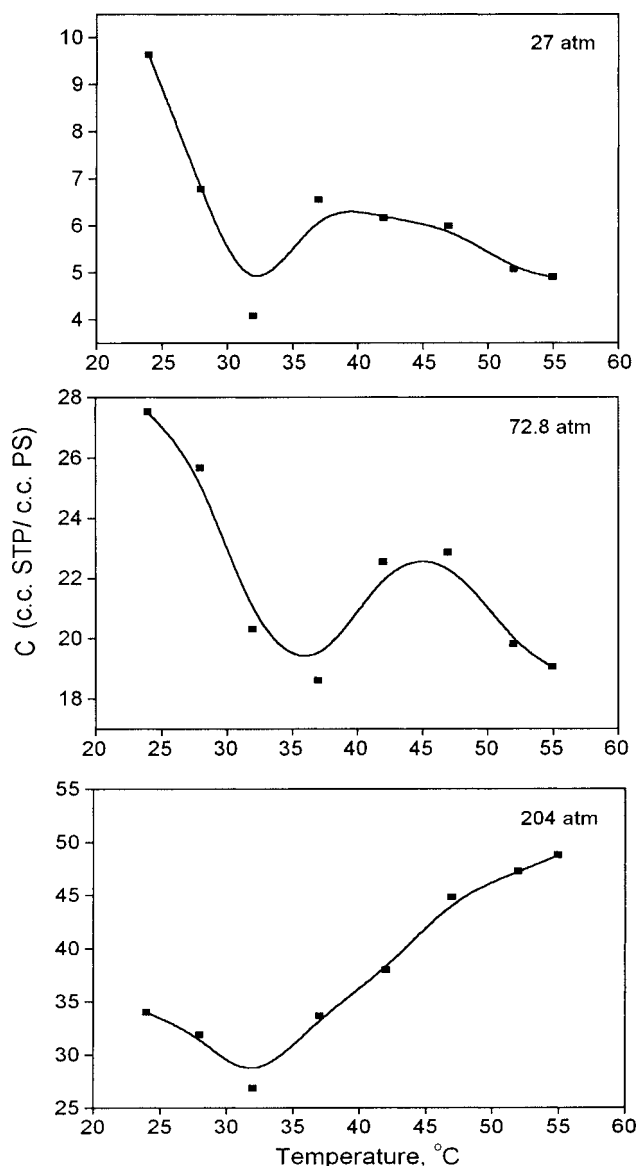


Figure 7. The CO₂ sorption isobars at 27, 72.8, and 204 atm for PS as a function of temperature.

Conclusions

The CO₂ sorption concentration (C) isotherms of glassy PMMA and PS as a function of pressure (P) consisted of two distinct regions separating at near the critical pressure for each polymer. The sorption isotherms in these two distinct regions could both be described fairly well by the power law $C = kP^n$, which functioned better than the Langmuir model or Henry's law. The sorption isobars of PMMA and PS provided information on the existence of chemisorption and physisorption. The temperature necessary for chemisorption to occur was lower for PMMA than for PS due to the presence of the specific interaction between the carbonyl groups of PMMA and CO₂ as compared with PS.

Acknowledgment

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